Economics for Engineers

UNIT-1

① Introduction :

(i) Economics Definition

(ii) Basic economic Problems

(iii) Resources constraints and welfare maximization

(iv) Micro and Macro Economics

(v) Production Possibility curve.

(vi) Circular flow of economic Activities

(vii) Basics of Demand

② Supply and Equilibrium –

(i) Demand side and supply side of the market

(ii) Factors affecting demand & supply

(iii) Elasticity of demand & supply – price, & income and cross-price elasticity.

(iv) Market equilibrium price

Marginal utility (consumer's equilibrium Market " Unit 2)

Introduction

① Economics :-

Economics is the study of how people, businesses, and governments use limited resources (like money, land, labor) to produce goods and services, and how these are distributed and consumed in society.

It mainly answers 3 basic questions :-

(i) What to produce ?
(ii) How to produce ?
(iii) Whom to produce ?

Economics = Study of resources, production, and consumption.

② Basic Economic Problems

→ Scarcity → Refers to the limitation of supply in relation to demand for a commodity.

- It means there are limited resources (like money, time, land, labor) but unlimited human wants.
- Resource are limited
- Human wants are unlimited.
- Because of scarcity, we cannot satisfy all needs.
- It forces people to make choices (what to buy, what to produce).
- Scarcity is the basic problem of economics.

— Because resources are limited and wants are unlimited every economy faces three main problems —

(i) What to produce ?

→ Out of so many goods and services, which one should be made?

Eg- Should land be used to grow wheat or cotton?

(ii) How to produce ?

→ Which method should be used to make goods-
more machines (capital- intensive)
or
more workers (labor- intensive).

Eg- should a factory use robots or hire more people?

(iii) For whom to produce ?

→ Who should get the goods and services produced — rich people, poor people or both?

Eg- Should more luxury cars be made for the rich, or more buses for everyone?

Microeconomics	Macroeconomics
① Study of individual units like a person, household, firm.	Study of the whole country (national or global).
② Tools- Demand & Supply	Tools- Aggregate demand & Agg. supply.
③ Also known as "Price theory."	Also known as "Income and Employment Theory"
④ Focuses on price of a commodity, demand & supply, production, costs, profit.	Focuses on National income, unemployment, inflation, economic growth.

Micro economics	Macroeconomics (3)
⑤ Eg- ~~No~~ Individual income, Individual output.	Eg- National Income, National output.
⑥ It is a small part of economy.	It refers to a whole economy.
⑦ It has limited scope	It has a wider scope.
⑧ It assumes all the macro variable constant. (national income, consumption, savings).	It assumes all the micro variables are constant (like decision of households and firms).

Resources constraints

Resources constraints ~~are~~ mean that resources like land, labor, capital, and raw materials are limited, while human wants are unlimited. Because of this economics face many challenges in meeting all needs.

Main challenge –

(i) Limited Production - Not enough resources to produce all goods and services.

Eg- A country with less oil cannot make enough petrol.

(ii) Need for choice - Gov. and businesses must ~~provide~~ decide what to produce and what to sacrifice.

Eg- A gov. may have to choose b/w investing in healthcare or defense due to budget limits.

(iii) Higher prices- When resources are short, prices of goods goes up.

Eg- Less wheat means costly bread.

(iv) Unemployment - Shortages of resources (like machines or skills) means fewer jobs.

Eg- Poor countries may not provide job to all workers

Welfare ~~or~~ Maximization (4)

Welfare- means well being, comfort, and happiness of people in society.

→ Welfare maximization means using the limited resources of a country in the best possible way so that the happiness, needs and well being of people are maximized.

- Resources should not be wasted, must be used efficiently.
- The aim is to give highest benefit to society.
- Wealth ~~so~~ and goods should be distributed fairly, not just among the rich

PRODUCTION POSSIBILITY CURVE (PPC)

chances, combination ; graph

Production possibility Frontier (PPF)
Transformation curve / Frontier / Boundaries

– It refers to the graphical presentation of possible combination of 2 goods which can be produced within given resources.

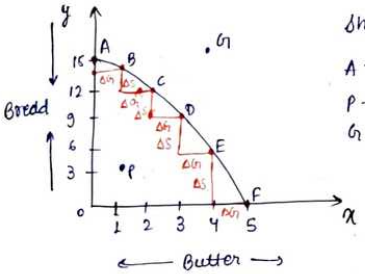
Assumptions → Technology is assumed to be constant.

- Resources are fixed.
- With given resources, only two goods can be produced.
- Resources are fully utilized

S = Sacrifice, Gn = Gain

TABLE

Combination	Bread	Butter	(Marginal rate of) Transformation MRT / MOC (Marginal opportunity cost) $\frac{\Delta S}{\Delta Gn}$
A	↓ 15	0 ↑	1Gn : 1S — 1/1 = 1
B	↓ 14	1 ↑	1Gn : 2S — 2/1 = 2
C	↓ 12	2 ↑	1Gn : 4S — 3/1 = 3
D	↓ 9	3 ↑	1Gn : 4S — 4/1 = 4
E	↓ 5	4 ↑	1Gn : 5S — 5/1 = 5
F	↓ 0	5 ↑	



Shape → Concave

A → Full utilization of Resources.

P → Underutilization

G → Unachievable.

• OPPORTUNITY COST – It is the value of next best alternative you give up when you make a choice.

Opportunity-cost = What you give up to get something else.

→ Every choice has a trade-off

→ It shows what you sacrifice to get something

→ Helps in making better economic decisions

Eg- If a farmer grows wheat instead of corn, the opportunity cost = profit from corn not grown.

Marginal ~~cost~~ opportunity cost (MOC)

→ Refers to the no. of units of a commodity sacrificed to gain one additional unit of another commodity.

Eg- If do produce 1 extra unit of butter, you must give up 2 units of bread → MOC = 2 breads.

$$MOC = \frac{\text{Units of goods sacrificed}}{\text{Units of goods gained}}$$

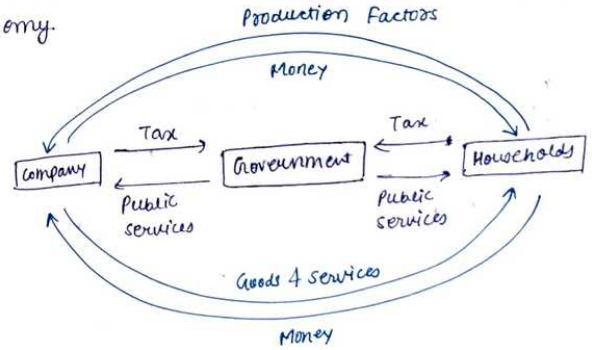
Marginal rate of Transformation-(MRT)

→ MRT is the rate at which one good is sacrificed to produced an extra unit of another good, keeping resources fully used.

$$MRT = \frac{\Delta \text{ Good A (sacrificed)}}{\Delta \text{ Good B (gained)}}$$

Circular flow of Economic Activities.

It is a model that shows how money, goods and services moves in a continuous loop b/w different sectors of the economy.



→ Household (people) give money to companies by buying goods and services.

→ Companies give goods and services to households in return.

→ Both households and companies pays taxes to the gov.

→ The gov. uses this tax money to provide public services (like roads, schools, hospitals) to households and companies.

Law of Demand & Supply

Condition — Homogeneous quality of goods
Large no. of buyers & sellers.

Law of Demand :-

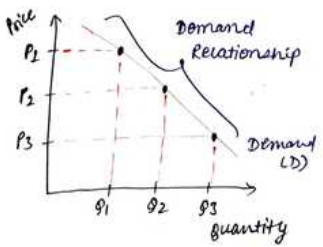
→ When the price of goods falls, people buy more of it.

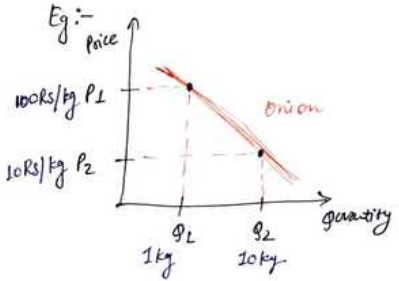
→ When the price rises, people buys less of it.

→ Because consumers always try to maximize satisfaction at lower cost

Price ↑ Price ↓
Demand ↓ Demand ↑.

[consumers perspective ⇓ talks about Affordability]





• Why law of demand always downward?

→ • Cheaper goods always replace costly ones. (Giffen goods)
• Lower price = more purchasing power
• Lower price attracts new buyers.
• Goods are used in more ways when cheaper.

Demand is downward because lower price increases quantity demanded.

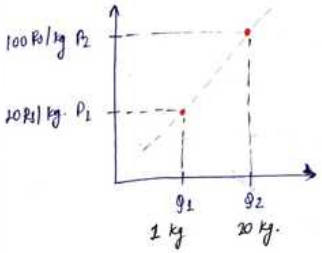
• Exception of Law of Demand :

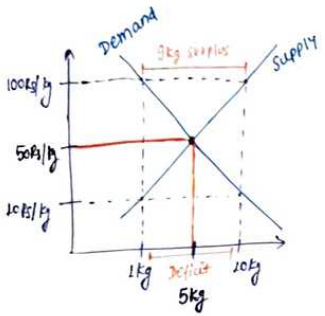
→ • Poor-quality items, people buy less when cheap.
• Higher price shows status, so people buy more.
• People wait if price may fall, or buy early if it may rise.
• Like salt / medicine, price change doesn't affect demand.

Law of Supply?

• When the price of good rises, producers sell more of it.
• When the price falls, producers sell less of it.
• Because sellers wants to earn more profit at higher prices.

Price ↑ Price ↓
Supply ↑ Supply ↓





Blue line – price 100 Rs/kg → demand (1kg)
पर supply करने वाला 10kg तो supply कर रहा। (9kg surplus)

- 10 Rs/kg है तो demand 10kg but ~~supply~~ supplier 1kg supply कर रहा है।

The midpoint is the Equilibrium point where
Demand = Supply

Equilibrium means a situation where demand and supply are equal at a certain price.

- Buyers wants to buy exactly the same ~~output~~ amount that sellers want to sell.
- There is no shortage and no surplus.

(2.) SUPPLY AND EQUILIBRIUM

Demand side and Supply side of the Market –

The demand side of the market shows how much buyers are willing to purchase at different price (price falls → demand rises). [demand curve slopes downward]

The supply side shows how much sellers are willing to sell at different prices (price rises → supply increases). (demand curve is ↗ upward)

The point where demand and supply are equal is called Equilibrium, where price is stable and quantity demanded = quantity supplied.

Factors affecting demand and supply :–

also known as Determinants of demand or demand function

① Price of commodity (P) :– P↑ D↓, P↓ D↑] negative relationship

② Price of related goods (P_r) :–

Cross demand → Substitute goods
→ complementary goods.

P_r
- Substitute goods → Pepsi & coke
  - P_{pepsi} ↑ D_{coke} ↑
  - P_{pepsi} ↓ D_{coke} ↓
  ] Rel^n +
- complementary goods → Phone & sim
  - P_{sim} ↓ D_{phone} ↑
  - P_{sim} ↑ D_{phone} ↓
  ] Rel^n –

③ Tastes and Preferences – Depends upon fashion, climate, religion, etc.

favourable T → D↑
Unfavourable T → D↓

Demand function :–

$g^d_x = f(P_x, P_y, I, T, O)$
- g^d_x → demand, product
- P_x → Price of commodity
- P_y → Price of related comm.
- I → Income
- T → Taste/preferences
- O → other.

Factors affecting Supply :-

Supply function

$S = f(P_x, C, T, E, N, G)$

- P_x → Price of the good
- C → Cost of Production (wages, raw materials, etc)
- T → Technology
- E → Future Expectations
- N → Number of sellers
- G → Government policies (taxes, subsidies, regulations).

- Price ↑ supply ↑
- Higher cost (wages, raw materials) → less supply.
- Better technology → more supply.
- If price expected to rise, suppliers may hold back

Elasticity of Demand and Supply

↓ Change

→ It shows how much demand or supply changes ~~the~~ when price, income, or related goods's price changes.

$$E_d = \frac{\%\text{ change in } q_d}{\%\text{ change in price}}$$

(q_d → quantity demand.)

$$= \frac{\frac{\Delta q}{q} \times 100}{\frac{\Delta P}{P} \times 100}$$

(Δ → change)

$\Delta q = q_1 - q_0$

$\Delta P = P_1 - P_0$

$P_0 = 10 \quad q_{d_0} = 200$

$P_1 = 20 \quad q_{d_1} = 100$

$\Delta q = 100 - 200 = -100$

$\Delta P = 20 - 10 = 10$

$$= \frac{\frac{-100}{200} \times 100}{\frac{10}{10} \times 100} = \frac{-50}{100} = -\frac{1}{2} = (-0.5)$$

-ve sign only shows the inverse Relationship b/w P & q_d.

~~Degree of Price~~

Elasticity

Price Elasticity

↳ Shows how much demand changes when the price of good changes.

Eg- If P_{tea} ↑, people may buy less tea.

→ If demand changes a lot → elastic

→ If demand changes little → inelastic

Income Elasticity

Shows how much demand changes when income of consumers changes

Eg- If your I ↑, you may buy more clothes or cars

→ For basic items like salt, rice, d does not change much

→ For cheaper goods, d ↓ as I ↑.

Cross Elasticity

Shows how demand of one good changes when the price of another good changes.

Eg- If P_{coffee} ↑, people may buy more tea (substitutes)

If P_{petrol} ↑, ~~peop~~ demand for cars may ↓ (complements)

Market Equilibrium

→ It is the price at which the quantity demanded = quantity supplied in the market.

- If price ↑ ⟶ surplus (more supply than demand)
- If price ↓ ⟶ shortage (more demand than supply).
- At Equilibrium → market is stable, no shortage or surplus.

Eg → If at ₹50, buyers want 100 units and sellers also supply 100 units → ₹50 is the equilibrium price.

UNIT-2

1. Theory of consumer :-

(i) Theory of utility and consumer's equilibrium

2. Indifference curve Analysis

3. Budget constraints

4. Consumer Equilibrium

5. Demand forecasting

(i) Regression Technique

(ii) Time - series

(iii) Smoothing Technique :-

(a) Exponential

(b) Moving Averages Method.

① Theory of consumer choice

→ It explains how consumers decide what to buy to get the maximum satisfaction from their limited income.

Theory of utility :-

Utility - Refers to the power or property of a commodity to satisfy human needs. (Satisfaction).

Two main approaches of to study consumer's behaviour.

① Cardinal utility Approach - Assumes that utility (satisfaction) can be measured in numbers.

② Ordinal utility approach - Assumes that utility cannot be measured, only ranked (1st choice, 2nd choice, etc.)

To understand the meaning of utility, there are 2 concepts :-

(i) Total ~~Utility~~ Utility and Marginal Utility.

(ii) Law of Diminishing Marginal Utility.

(i) # Total utility

→ Total satisfaction or happiness a consumer gets from consuming a certain quantity of a good or service.

Eg- Eating 3 pizza slices gives a total satisfaction of 30 utils → TU = 30.
1st slice → 20 utils, 2nd slice → 15 utils, 3rd slice → 10 utils

→ TU increases as consumption increases, but eventually may level off or decreases.

$TU_n = U_1 + U_2 + U_3 + \dots U_n$

$TU_n = 20 + 15 + 10 = 45.$

Marginal Utility

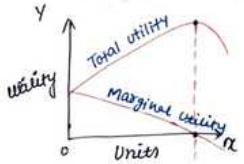
→ Additional satisfaction gained from consuming one extra unit of a good or service

Eg- Slices of Pizza (Q)	TU	MU
1	15	15
2	25	25-15 = 10
3	30	30-25 = 5.

$MU_n = TU_n - TU_{n-1}$

$MU_2 = MU_2 - T_1 = 25 - 15 = 10$ utils.

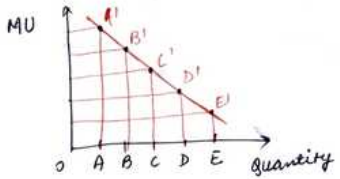
→ MU decreases as more slices are eaten → shows Law of Diminishing MU.



(ii) Law of Diminishing MU.

→ When you consume more and more of a good, the extra hapiness (satisfaction) you get from each additional units keeps decreasing.

Eg- First slice of pizza feels great, second less, third even less.



Consumer's Equilibrium

Relationship between TU & MU

(i) TU starts from origin / ↑ at diminishing rate / reaches top / starts falling.

(ii) MU starts from Top / falling (DMU) / becomes 0 / then -ve.

(iii) When TU is rising / MU is falling but +ve.

(iv) When TU is at top / MU is 0.

(v) When TU is falling / MU is -ve.

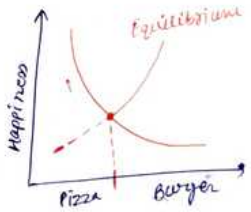
Consumer's Equilibrium :- It refers to that situation when a consumer is having maximum satisfaction with limited income and has no tendency to change his way of existing expenditure.

C·E under two different situations

- consumer spends his entire income on single commodity.
- Consumer spends his entire income on Two commodity.

Eg :-

- You like pizza and Burgers
- You spend your money so that the happiness / $ from pizza is the same as from burgers.
- If its equal, you're happy and don't want to change — that's consumer equilibrium.



Approaches

(i) Cardinal Utility Approach (Two commodity)

- Assumes utility can be measured in numbers (utils).
- Condition for equilibrium -

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y} = \dots = \frac{MU_n}{P_n}$$

where, MU_x = MU of good x.

P_x = Price of good x.

Eg- price of pizza = ₹50, Burger = ₹25

MU of last pizza slice = 100 utils, burger = 50 utils

$$\frac{MU_{pizza}}{P_{pizza}} = \frac{100}{50} = 2, \quad \frac{MU_{burger}}{P_{burger}} = \frac{50}{25} = 2.$$

- Consumer is in equilibrium.

Equilibrium condition of single commodity.

→ $(MU_x) = (P_x)$, paid for the commodity.

Units of x	P_x	MU (utils)	Remarks
1	10	20	$MU_x > P_x$ consumer will increase the consumption
2	10	16	
3	10	10	$MU_x = P_x$ (C·E)
4	10	4	$MU_x < P_x$ decrease the consumption
5	10	0	
6	10	-6	

~~Budget constraints~~

Ordinal Utility Approach :-

- Assumes that utility cannot be measured numerically, but goods can be ranked in order of preference.

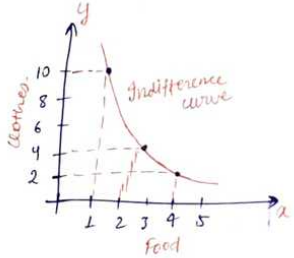
Indifference Curve Analysis, Budget constraints comes under ordinal approach.

Indifference Curve Analysis :-

→ A curve showing all combinations of two goods that give a consumer the same level of satisfaction.

Eg- You are equally happy with:

- 10 units of clothes + 1 unit of food
- 4 units of clothes + 2 unit of food.
- 2 units of clothes + 4 units of food.



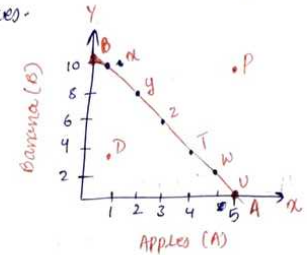
- If you get more food, you need less clothes to stay equally happy.
- Curve slopes downward, showing the trade-off between food and clothes.

* Marginal Rate of Substitution (MRS)

→ Shows how much of one good a consumer is willing to give up to get on more unit of another good, without changing their happiness.

Budget constraints.

- It shows all the combination of two goods a consumer can buy with their income and given prices.
- Think of it as your "shopping list" based on money and prices.



P = Unattainable combination

D = income doesn't spend all.

→ all income spent

- The line shows budget line. (all the combos I can afford).
- Point inside the line = you don't spend all income.
- Point outside = cannot afford.

→ It basically represents the limit on consumption because a consumer cannot spend more than their income.

For two goods X and Y.

$$P_x \cdot X + P_y \cdot Y \leq I$$

- P_x : Price of good x
- X : quantity of good X
- P_y : Price of good Y
- Y : quantity of good Y
- I : Income

Slope of budget line $= -\frac{P_x}{P_y}$

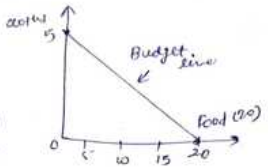
Eg- Suppose,

Income I = ₹1000

P_F (food) = ₹50 per unit

P_C (clothes) = ₹200 / unit

Budget eqⁿ $50F + 200C = 1000$



If consumer's spend all income on

Food only $\frac{1000}{50}$ = 20 unit

clothes only = $\frac{1000}{200}$ = 5 units.

So, the budget line connects these two points.

(Food 20, clothes 0) and (Food 0, clothes 5).

Demand forecasting.

→ Demand forecasting means predicting how much of a product or service people will want in the future by looking at past data, trends and other factors like seasons, prices, or customer habits.

→ It helps businesses to plan production, manage stock, set prices, and avoid shortage or waste.

- Regression Technique

→ Used when demand depends on certain factors (called independent variables).

Eg- Ice cream sales (demand) depend on temperature, price, and income levels.

→ Regression finds a mathematical relationship (eqⁿ) b/w demand and these factors.

- Types

→ Simple Regression (demand vs one factor eg- price)

→ Multiple Regression (demand vs many factors)

Estimating Total Cost Through Regression –

Month	Units Reduced X	Total Cost Y	X^2	Y^2	XY
Jan	10000	40	100	1600	400
Feb	15000	50	225	2500	750
Mar	20000	60	400	3600	1200
Apr	25000	70	625	4900	17500
May	30000	80	900	6400	2400
	$\Sigma X = 100$	$\Sigma Y = 300$	$\Sigma X^2 = 2250$	$\Sigma Y^2 = 19000$	$\Sigma XY = 6500$

Calculate total cost of 18 units

$Y = a + bX$

$\boxed{Y = 20 + 2X}$

$Y = 20 + 2(18)$

$Y = 20 + 36$

$Y = 56000$

Formula

$$b = \frac{n\Sigma XY - \Sigma X \Sigma Y}{n\Sigma X^2 - (\Sigma X)^2}$$

$$a = \frac{\Sigma Y}{n} - b\frac{\Sigma X}{n}$$

$Y = a + bX$

$$b = \frac{5(6500) - (100)(300)}{5(2250) - (100)^2}$$

$\boxed{b = 2}$

$a = \frac{300}{5} - 2\left(\frac{100}{5}\right)$

$\boxed{a = 20}$.

Time Series

- Assumes demand changes with time in a predictable pattern.
- Past demand data is used to forecast the future. Looks at trends, seasonal effects, and cycles.
- Eg- Sales ~~with~~ rise every december due to festivals.
- A Time series is a set of data collected or recorded at different points of time, usually at equal intervals (daily, monthly, yearly).

components :-

① Trend (T) - long-term movement (upward or downward) in data over a long period.

Eg- ↑ demand for phones over years.

② Seasonal (S) - Regular patterns that repeat every year or period.

Cold drink sales increase in ~~smalls~~ summer.

② Cyclical (C) - Car sales go down in recession years.

Sales drop suddenly due to floods,

④ Irregular / Random (I) → strikes or pandemic.

Smoothing Techniques.

→ Means removing random fluctuations (ups and down) from data to see the real trend more clearly.

→ It is used in demand forecasting and time series analysis to make future predictions more accurate.

Exponential Smoothing

A smarter method that gives more importance to recent data and less to older data.

Uses a number called smoothing constant (α) between 0 and 1.

$$F_{t+1} = \alpha D_t + (1-\alpha) F_t$$

- F_{t+1} : Forecast of next period
- α : smoothing factor
- D_t : Actual demand in this period
- F_t : forecast of this period.

If α is high → recent data matters more

$\alpha \downarrow$ → past forecast matter more

$\alpha = 0.1$

Month	Demand	F_t*	$\alpha D_t + (1-\alpha) F_t$
1	46	—	
2	56	46	$0.1(46) + (1-0.1)(46)$
3	54	47	$0.1(56) + 0.9(46)$
4	43	47.7	$0.1(54) + 0.9(47)$
5		47.23	$0.1(43) + 0.9(47.7)$

Find — got the forecasted value

Moving Average

→ You take the average of demand / sales from the past few periods to forecast the next period.

It helps remove ups and down in data.

Eg- • sales for Jan, feb, March = 100, 120, 140

• Moving Avg (3 months) = $\frac{(100+120+140)}{3} = 120$

• So, forecast for April = 120

UNIT - 4

National Income Accounting

→ Money value of all the final goods and services produced by a country during one financial year

→ One financial year – 1st April and ends on 31st March.

→ It includes wages, interest, rent, profit, received by factors of production like labour, capital, land and entrepreneurship of a nation

* National Income – It is the total ~~year~~ income earned by all people of a country from all economic activities during one year.

• Helpful in determining the progress of the country.

National Income formula = Consumption + Government Expenditures + Investments + Exports – Imports + Foreign Production by National Residents – Domestic Production by Non-National Residents.

Factor cost, Basic Price, Market Price

Laptop Manufactoring – ~~15000~~

Raw material – 15000

Land (Rent) – ~~2000~~ 1000

Labour (wages / salary) – 2000

Capital (Interest) – 2000

Total = ₹2000 → (Factor cost)

→ Factor cost is the amount paid to factors of production (labour, land, capital, entrepreneur) for producing goods and services.

• Basic Price

→ The price received by the producer for goods and services.

→ It includes subsidies but excludes taxes on products.

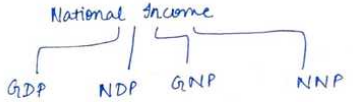
· Basic Price = Factor cost + Production Taxes - Production Subsidy

• Market Price -

→ The price paid by the consumer in the market.

→ It includes taxes and excludes subsidies.

Market Price = Basic Price + Product Taxes - Product Subsidy



① GDP - Total value of all goods and services produced inside the country in one financial year.

→ Gross Domestic Product.

→ Refers to the value of goods produced within the geographical territory of a country irrespective of whether they are produced by citizens and foreigns.

Types of GDP

① Real GDP → Value of all goods & services used in calculating GDP at Base year / Price.

② Nominal GDP

→ The value of goods and services used in calculating GDP at the current price

② GNP (Gross National Product) :-

→ GDP + Net income from abroad.

→ Measures income earned by country's people anywhere in the world.

③ Net Domestic Product (NDP)

→ GDP - Depreciation (wear and tear of machines).

→ Shows actual production after replacing used capital.

④ NNP (Net National Product) -

→ GNP - Depreciation.

→ True measure of National Income.

⑤ NI (National Income) -

→ NNP at factor cost

→ Total income earned by factor of production (wages, rent, interest, profit)

⑥ Personal Income (PI) -

→ Income actually received by individuals

→ Includes salaries, pensions, scholarships, etc.

⑦ Disposable Income (DI) -

→ Personal Income - Direct taxes

→ Income available for spending and saving.

Overview of Macroeconomics

→ Macroeconomics is the branch of economics that studies economy as a whole.

→ It looks at overall economic performance rather than economics individual people or firms.

① National Income -

Measures total income and output of a country (GDP, GNP).

② Economic Growth -

Increase in country's production and income over time.

③ Inflation –

Rise in general price level of goods and services.

④ Unemployment –

→ No. of people who are willing to work but cannot find jobs.

⑤ Money and Banking.

→ Role of money, bank, and central Bank (RBI).

⑥ Fiscal Policy

→ Gov. spending and taxation to control the economy.

Monetary Policy –

→ Control of money supply and interest rates by central bank.

OBJECTIVES OF MICROECONOMICS.

(i) Economic growth
(ii) Price stability
(iii) Full employment
(iv) Economic stability.

Microeconomics Issues :–

① Inflation –
- Continuous rise in general price level of goods and services.
- Reduces purchasing power of people.

② Unemployment –

③ Economic growth

④ Poverty – Large section of people unable to meet basic needs like food, shelter, and clothing

⑤ Income Inequality – Unequal distribution of income among people.

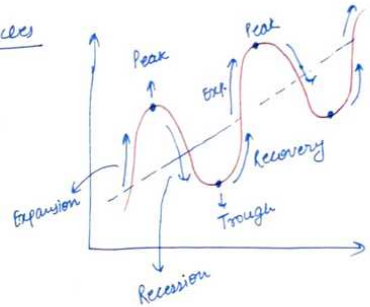
⑥ Balance of Payment (BOP) problems –
- When imports are more than exports, leading to deficit
  ↓
  spending more than you earn

⑦ Fiscal Deficit
→ Gov. expenditure exceeds gov. revenue.

⑧ Public Debt
→ High borrowing by the gov.

Business Cycles

- Refers to the fluctuations (ups and down) in the level of economic activity of a country over a period of time.
- These changes affect production, income, employment, prices and investment.



A business cycle explains why an economy sometimes grows faster and sometimes slows down or falls.

PHASES – Expansion, Peak, Contraction, Trough, Recovery.

① Expansion (Boom)
→ When economic activity increases continuously.
→ Increase in production and income.
→ High employment
→ Rise in investment and profit.
→ Business confidence is high

② Peak
→ Where economy reaches the highest level of economic activity.
→ Maximum production and employment.
→ Economy works at full capacity.
→ Prices start rising.
→ No further growth possible

③ Contraction (Recession) –
→ It is the phase when economic activities starts declining.
→ Fall in production and income.
→ ↑ in unemployment.
→ ↓ in Demand and investment
→ Business confidence falls.

④ Trough (Depression) –
→ It is the phrase where the economy reaches its lowest level of economic activity.
→ Very low production and income.
→ High Unemployment.
→ Low demand and investment.
→ Economy starts preparing for recovery.

⑤ Recovery
→ Stage after depression (trough) when economy starts improving slowly.
→ Increase in production
→ Rise in employment
→ ↑ in demand.
→ ↑ in investment.
→ Improvement in income

Causes of Business Cycle

① Changes in Demand
② Changes in investment.
③ Govt. policies.
④ External shocks (pandemic, wars).

Inflation
— Situation where the general price level of ~~output~~ goods and services keeps rising over time; which reduces the purchasing power of money.

Types

① Demand-Pull Inflation
— caused when demand is more than supply
— Prices rises because people want to buy more goods than available.

② Cost-Push Inflation
— Caused by increased in cost of Production
— Rise in wages, raw materials, or taxes ↑ prices.

③ Creeping Mid Inflation
— Prices rise slowly.
— Considered normal and manageable

④ Walking Inflation
— Prices rises at a moderate rate.

Causes

① Demand-Pull Inflation ② Cost-Push Inflation
③ Increase in money supply.
→ Too much money in circulation.
→ Leads to higher spending and higher prices.

Consequences (Effects) to Inflation:-

* positive effect
(i) Higher profits for producers
(ii) Encourages investment (moderate inflation)

Negative effects

1. ↓ purchasing power of money.
2. ↑ inequality
3. Creates economic instability.

COST Theory and Analysis

Cost:- Cost means the money spent by a firm to produce goods or services.

Cost is the total expenditure made on inputs or resources that are used for the production of final goods or services.

Cost = Explicit cost + Implicit cost

1. **Explicit cost** → Refers to those costs which involve direct cash payment by the firm.
   - It involves actual money payment on buying and hiring inputs
   - Eg- Rent paid for factory / shop, wages paid for workers, cost of raw materials, Electricity bills etc
2. **Implicit cost** → Implicit cost refers to the cost of self-owned resources used in the business, for which no direct cash payment is made, but they are opportunity cost of self-owned resources.
   - Eg- Rent of own land.
   - A person owns a shop:
   - He works himself but takes no salary.
   - If he worked somewhere else, he could earn 20,000.
   - 20000 is implicit cost.

Explicit cost	Implicit cost
1. Involves cash payment.	1. No direct cash payment
2. Also called out-of-pocket cost.	2. Also called opportunity cost.
3. Money is actually paid by the firm	3. Money is not paid, but income is sacrificed.
4. Eg- Rent, wages, raw materials etc	4. Eg- owner's salary, own building rent etc

TYPES – ① Fixed cost ② Variable cost ③ ~~Ad~~ Total cost.

Cost function – Relation b/w cost and output.

$C = f(q)$

C → cost, f → functional relationship, q → quantity.

Short run is a time period in which atleast one factor of production is fixed
- Some cost are fixed
- Some cost are variable

SHORT RUN COST
↙ ↘
Short Run Fixed cost + SR Variable cost

~~③ Fixed cost~~

Total cost → Fixed cost + Variable cost

COSTS
↙ ↓ ↘
Total cost | Average cost | Marginal cost.

① SR Fixed cost –
cost which does not change with the level of output.
Even if production is zero, this cost must be paid.

Eg –
- Factory rent
- Salary of permanent staff
- Insurance

- If rent is ₹10,000 per month, it remains ₹10,000 whether output is 0 or 100 units.

② SR Variable cost → Variable cost changes with production.
More production → more variable cost

Eg → Raw material
- Wages of daily workers.
- electricity (usage based)

- Producing more shirts needs more cloth → variable cost increases.

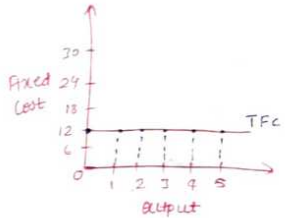
TC → TFC, TVC, TC

A: TOTAL FIXED COSTS – (TFC)

→ Total cost that does not change with the level of output.

→ These cost are fixed in nature, which means :–
- They remain the same at all levels of output.
- Exists even at zero output.

Eg – Rent of factory
Insurance etc.



Output	TFC
0	12
1	12
2	12
3	12
4	12
5	12

A firm pays ₹20,000 per month as rent and salary of permanent staff. Whether it produces 0 units, 50 units, or 200 units, the cost remains ₹20,000 (↓ TFC).

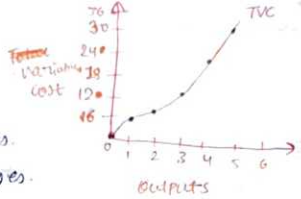
Shape of TFC curve :–
→ The TFC curve is a straight horizontal line.
→ It is parallel to the X-axis.

B. Total Variable cost :- (TVC)

→ It is the total cost that changes with the level of output.

→ These cost are called variable because :

- They increase when output increases.
- They decrease when output decreases.
- They are zero when output is zero.



Eg- Cost of raw materials,
Wages of daily workers,
Electricity (usage based)

Output	TVC
0	0
1	6
2	10
3	15
4	24
5	35

Shape of TVC curve :-

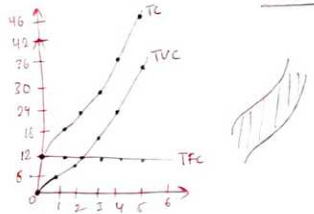
→ The curve is inverse S-shaped.

→ The TVC curve starts from the origin

→ Initially, it rises slowly,

→ After some level of output, it rises rapidly.

Total Cost



Output	TFC	TVC	TFC + TVC = TC
0	12	0	12+0 = 12
1	12	6	12+6 = 18
2	12	10	12+10 = 22
3	12	15	12+15 = 27
4	12	24	12+24 = 36
5	12	35	12+35 = 47

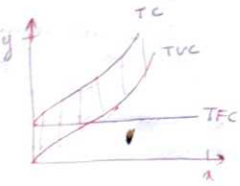
- Total expenditure incurred by a firm to produce a given level of output.

$TC = TFC + TVC$.

Shape of TC curve :-

- At zero output, TVC = 0 ⇒ TC = TFC at 0 output. TC curve starts from TFC level, not from origin.
- TC and TVC are parallel to each other.

Relationship b/w TFC / TVC / TC



① Output → X Axis
Costs → Y axis

② Shape
TC / TVC = Inversely S shaped.
TFC = Horizontal straight line parallel to x axis.

③ At 0 level of output
TFC = TC, there is no VC.

④ TC / TFC do not start from zero / origin.

⑤ Distance between TC & TVC are same
" " TC & TFC is called TVC

Average Costs

- Average cost (AC) is the cost of producing one unit of output.
- On average, how much does it cost to produce one unit.

$AC = \frac{TC}{Q}$ → Total cost
Q → output (quantity)

जैसे output बढ़ता रहता है, Avg cost कम होता जाता है।

AC → ATC
Avg Total cost
AFC — Avg fixed cost
AVC — Avg variable cost

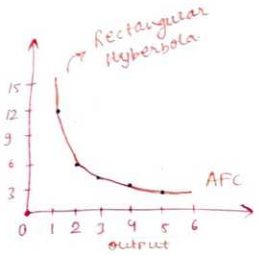
TFC → $AFC = \frac{TFC}{Q}$

TVC → $AVC = \frac{TVC}{Q}$

TC → $AC = \frac{TC}{Q}$ or AFC + AVC.

① Average Fixed cost :-

Output	TFC	$AFC = \frac{TFC}{Q}$
0	12	undefined.
1	12	12
2	12	6
3	12	4
4	12	3
5	12	2.4
6	12	



AFC curve –

→ AFC curve slop downwards as AFC falls with increase in output.

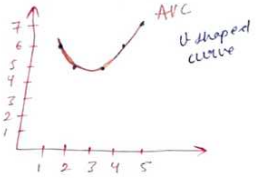
→ AFC is a rectangular hyperbola.

② # AVC Avg Variable cost

→ per unit variable cost of production

$$AVC = \frac{TVC}{Q}$$

output	TVC	AVC
0	0	—
1	6	6
2	10	5
3	15	5
4	24	6
5	35	7



AVC curve – constant,

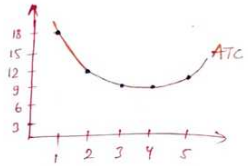
- First decreases, then increases.
- U-shaped curve

③ Average Total cost (ATC)

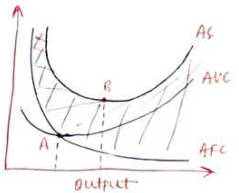
→ per unit total cost of production.

AFC + AVC = AC

output	AFC	AVC	AC
0	∞	—	—
1	12	6	18
2	6	5	11
3	4	5	9
4	3	6	9
5	2.40	7	9.40



Relationship b/w AFC / AVC / AC



① O/P → X axis, VC → Y axis.
② AFC → Rectangular hyperbola
③ AVC/AC → U-shaped curve ~~because~~
④ Distance b/w AC & AFC keeps on Rising.
⑤ Distance b/w AC & AVC keeps on Reducing. but they will never intersect because of AFC.

MARGINAL COST –

Refers to addition to total cost when one more unit of output is produced.

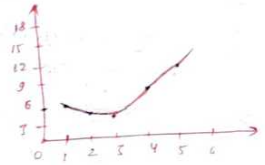
$$MC = \frac{\text{Change in Total cost}}{\text{change in units of output}} = \frac{\Delta TC}{\Delta Q}$$

10 → ₹50 ⎫ 55 − 50 = 5
11 → ₹55 ⎭ $\frac{5}{1} = 5$

$$MC_n = TC_n - TC_{n-1}$$

TC change होगा variable cost (VC) की वजह से, MC change होता है TC की वजह से, ultimately MC, change होगा VC की वजह से।

output	TVC	TFC	TC	MC $TC_n - TC_{n-1} = MC_n$	MC $TVC_n - TVC_{n-1} = MC_n$
0	0	12	12	—	—
1	6	12	18	18−12 = 6	6−0 = 6
2	10	12	22	22−18 = 4	10−6 = 4
3	15	12	27	27−22 = 5	15−10 = 5
4	24	12	36	9	9
5	35	12	47	11	11

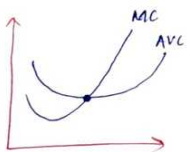


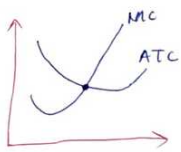
$$MC_n = TC_n - TC_{n-1}$$

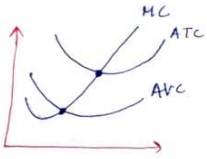
n = No. of units produced
MC_n = Marginal cost of nth unit
TC_n = Total cost of n units
TC_{n-1} = Total cost of (n−1) units.

Relationship
① ⁂ MC & AVC
② MC & ATC
③ MC & AVC & ATC

① MC will cut AVC at its minimum.
② MC ↓ AVC ↓ AC ↓
③ MC ↑ AVC ↑ AC ↑

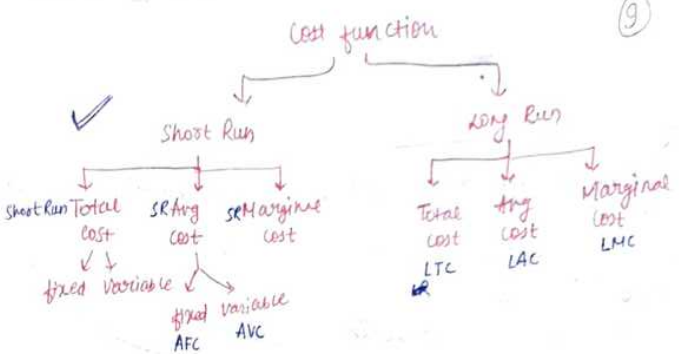






Fixed cost	Variable cost
① Does not change with increase and decrease in output.	① Cost changes with increase or decrease in output.
② FC exists even when output is zero.	② VC zero when output is zero.
③ FC is related to fixed factors like land, ~~land~~, building, machinery.	③ VC is related to variable factors like labor and raw material.
④ Also known as indirect cost.	④ Also known as direct cost.
⑤ Eg- salary of staff, Rent, insurance etc.	⑤ Eg- power, fuel, raw materials etc.

Total cost	Average Cost	Marginal cost
① Total expenditure incurred to produce a given level of output.	Cost of producing one unit of output.	Extra cost of producing one additional unit.
② Includes both fixed cost and variable cost.	Includes Avg fixed cost and Avg variable cost.	Depends only on change in variable cost.
③ $TC = TFC + TVC$	$AC = \frac{TC}{q \text{ (output)}}$	$MC = \frac{\Delta TC}{\Delta q \text{ (outputs)}}$
④ TC curve starts above the origin.	AC curve is U-shaped.	MC curve cuts AC at its lowest point.



LONG RUN

→ The long run is a period of time in which all factors of production are variable.

→ In the long run, a firm can:
- Change Plant size
- Buy new machines
- Expand and reduce scale of production.

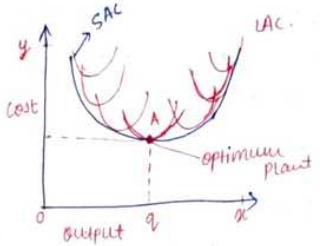
→ There is no fixed cost in the long run.

① Long Run Average cost (LAC)

→ Refers to the cost per unit of output in the long run, when all factors of production ~~of~~ are variable and the firm can change ~~of~~ its plant size.

→ LAC is generally U-shaped (flatter than short run avg cost curve)

→ U-shape of the LAC curve implies that the LAC falls first, reaches the minimum and then rises.



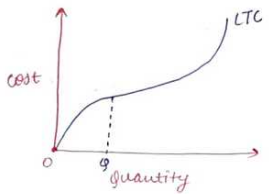
Optimum Output :-

→ Point q is the minimum point of LAC.
- Lowest cost per unit.
- Best plant size.
- This plant is called the optimum plant

At this point SAC = LAC (minimum).

Long Run Total cost (LTC):-

- LTC curve is ~~drawn~~ derived from short-run total cost (STC).
- LTC - shape of LTC curve is an inverse 'S' shape.
- LTC increases with increase in output.
- LTC shows the total cost of production when the firm has full freedom to adjust all inputs (change its plant size).



★ $LTC = LAC \times Q$

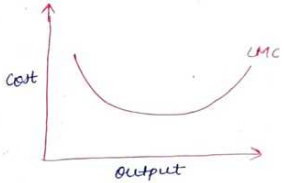
- LTC increases slowly at first and then increases rapidly after a certain level of output.

Long Run Marginal cost (LMC):-

- LMC shows how total cost changes when output is increased by one unit in the long run.

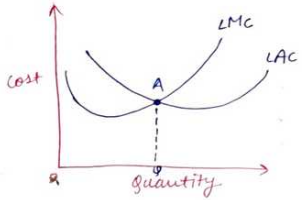
$$LMC = \frac{\Delta LTC}{\Delta Q}$$

- LMC curve is U-shaped.
- U-shaped of the LMC curve will be flatter than SMC curve.



Relationship b/w LAC & LMC:-

- LMC cuts LAC at its minimum point.
  - ↳ LMC < LAC → LAC falls.
  - ↳ LMC > LAC → LAC rises.



Short Run Cost	Long Run Cost
① Short run is a period in which at least one factor is fixed.	① Long run is a period in which all factors are variable.
② Fixed cost exists in short run.	② Fixed cost does not exist in long run.
③ Include both fixed and variable costs.	③ Includes only variable costs.
④ Plant size cannot be changed.	④ Plant size can be changed.
⑤ cost depend on a given plant size.	⑤ cost depend on best plant size.
⑥ Short-run cost curves are TFC, TVC, TC, AFC, AVC, AC, MC.	⑥ Long-run cost curves are LTC, LAC, LMC.

★ Short Run cost function
→ Shows the relationship between output and cost when atleast one factor of production is fixed.

★ Long Run cost function
→ Shows the relationship between output and cost when all factors of production are variable.

Economies of Scale

→ When the cost per unit of production (Average cost) decreases because the output (sales) increases.

→ Bigger production = lower cost per item.

Eg- If a factory makes 10 notebooks, the cost per notebook is high. If the same factory makes 10,000 notebooks, the machines, electricity, and workers are used better, so each notebook costs less.

Why economies of scale occur :-

(i) Machines are used more efficiently
(ii) Workers become faster and better at their work
(iii) Raw material are bought in bulk at cheaper prices.

Types of Economies of Scale

① Internal Economies of scale -

→ Cost advantages a firm gets when it increases its own size of production.

→ As the firm grows bigger, it can use better machines, skilled workers, bulk buying, and better management, which reduces the cost per unit of production

- Technical economies - Big firm use better machines.
- Managerial economies - work is managed better in big firms.
- Financial economies - Big firm gets money more easily and cheaply
- Risk-bearing Economies - Big firm can handle losses better.

② External Economies of scale -

→ Cost advantages that all firms in an industry get due to the growth of the entire industry.

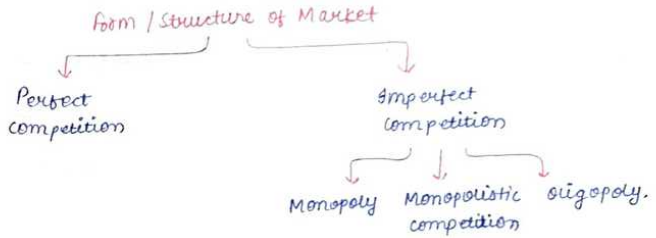
→ When many firm grows together, they get better roads, skilled workers, nearby suppliers, and shared facilities, which reduces cost for all firms.

Internal - Benefit from own growth
External - Benefit from industry growth

Market structure

- Market :- A market is a place or system where buyers and sellers come together to buy and sell goods and services at a price.

A market is any situation where buying and selling happens.



- Perfect competition -

→ When there is a large number of sellers and buyers of a commodity, and no individual buyer or seller has any control over its price

→ It is a market where everyone sells the same product at the same price, and no seller is powerful enough to change the price

Eg - Agricultural market (wheat, rice)

Features

① Large No. of Buyers and sellers

→ Large No. of Buyers and sellers, so no one can control the market

② Homogeneous Product

→ All firms sell identical products. (no difference in quality)

③ Free entry and exit -

→ Firms can enter or leave the market easily.

④ Price Taker
→ Firms accept the price fixed by demand and supply.

⑤ Perfect Information
→ Buyers and sellers knows everything about prices and products.

⑥ No advertising cost
→ Since products are same, no need for advertisements.

Degree of competition :-
- It shows how much competition exists among firms in a market
- Ranges from very high competition to no competition.

Degree of competition in perfect competition :- Very High

Monopoly :-
→ Monopoly is a market structure where a single seller sells a product with no close substitutes and has full control over price.

Mono → Single
Poly → firm control

Eg - Indian Railway, Electric supply in a city.

Features of Monopoly.

① Single seller -
→ Only one firm controls the entire market.

② No close substitutes -
→ consumers have no alternative choice

③ Price Maker -
→ The monopolist can fix the price of the product.

④ High barriers to entry -
→ other firms cannot enter due to legal, technical, or financial barriers.

⑤ control over supply
→ The sellers controls the quantity supplied in the market.

Degree of competition : No competition.

Monopolistic competition -
→ It is a market structure where many sellers sell similar but not identical products.

→ Each firm has some control over price, but competition still present.

→ Monopolistic → Monopoly + Perfect competition.

→ Degree of competition - Moderate competition.

→ Eg → Fast food outlets, cosmetic brands, Mobile phone brands etc.
↓
Many restaurants sell similar food but with different taste, price and service and customers can easily choose another restaurants.

Features

① Large no. of sellers - Many firm operate in the market

② Product differentiation - Products are similar but different in brand, quality, design, taste are packaging.
Eg - Toothpaste brands, restaurants etc.

③ Some control over price -
→ Firms can slightly increase and decrease prices.

④ Free entry and exit -
→ New firm can enter and old firms can leave easily.

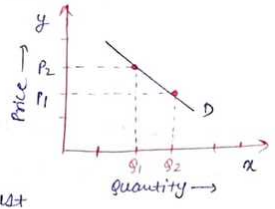
⑤ Advertising -
→ Firm advertise to show how their product is different

⑥ close substitutes -
→ Many alternatives are available for buyers.

~~# Oligopoly~~

Demand curve for a Monopoly Firm :-

- Full control over price under Monopoly.
- But doesn't mean the monopolist can sell any amount of commodity at any price.
- To sell more units, the monopolist must reduce the price.
- Demand curve slopes downward



Oligopoly :-

- Form of market in which there are a few big firms and a large number of buyer of a commodity.
- Each firm's decision affect to others, so competition is limited. Eg- Telecommunication (Jio, Airtel, Idea) major layer.

Features :-

(1) Few sellers - only a smaller no. of firms control most of the market.

(2) Interdependence - Each firm depends on the actions of other firms (price, output, offers).

(3) Product Type - Product may be same (cement, steel) or different (cars, mobile phones).

(4) High entry barriers - New firms find it difficult to enter due to high cost, technology, or brand power.

(5) Non-price competition - Firms compete through advertising, quality, discounts, and services rather than price.

(6) Possibility of collusion → Firms may secretly agree to fix prices or output (cartel).

Types of oligopoly :-

(1) Pure (Perfect) Oligopoly -
- Firms sell identical products.
- No product difference.
- competition is mainly through price.

Eg - Cement, Steel.

(2) Differentiated oligopoly -
- Firms sell similar but different products.
- Competition through advertising and branding.

Eg - cars, Mobilephones.

(3) open oligopoly -
- New firms can enter the market.
- Entry barriers are low or moderate.

(4) closed oligopoly -
- No new firm can enter.
- High entry barriers.

Eg - Telecom sector.

(5) collusive oligopoly -
- Firms work together instead of competing.
- They make joint decisions about price, output, market sharing.

Eg - ~~But~~ Oil producing countries.

(6) Non-collusive Oligopoly -
- Firms do not ~~compete~~ cooperate.
- Each firm takes decisions independently, but always keeps rivals (competitors) in mind.

Eg - Telecom companies,
Automobile companies etc.

~~11 National Income Accounting~~

• Differentiation

Feature	Perfect competition	Monopolistic competition	Monopoly ~~competition~~	Oligopoly
No. of Sellers	Large	Large	one seller, but large no. of buyers.	A small no. of big sellers and a large no. of buyers.
Type of Product	Homogeneous / Identical	Product differentiation	Unique (as there are no close substitutes of the monopoly product)	Homogeneous or differentiated.
Price control	None (Price taker)	Limited	Full (Price maker)	~~Not Uniform because of product differentiation.~~ Partial control
Entry & Exit	Free	Free	Very Hard / Blocked	Hard / Restricted.
Knowledge of Market conditions	Perfectly knowledge	Imperfect knowledge	Imperfect knowledge	Imperfect knowledge
Advertising	Not needed	Important	Rare	Often used (Non-price competition)